

exercise-4-recsys-2024

April 14, 2024

1 Item-item Collaborative Filtering

RecSys Exercise 4

1.1 Overview

In this exercise you will explore item-based collaborative filtering.

Please upload in Moodle this notebook (executable and in PDF) with the code and answers to the posted questions.

1.2 Instructions

Before you start just to remind the intuitive algorithm of the item-item collaborative filtering:

1. Compute similarity between pairs of items (cosine similarity).
2. Isolate the set of the most similar items based on the similarity measures. (correlation values: 1, the most similar user to -1 , the least similar user).
3. Compute the prediction score on an item i (that a user u has not rated) by adding the ratings $r(i, j)$ given by the user u on items similar to i . Each rating is weighted by the corresponding similarity $s(i, j)$ between items i and j .
4. Recommend to the user the items with the highest prediction scores.

2 Task 1: Compute item-item similarity

For this task please use the dataset named `movie_ratings.csv`. You can compare the dataset to a 20x20 matrix where columns represent users, rows represent movies, and each cell represents a user-movie rating (a value of 0 means that the user has not rated that movie). Note that the library `sklearn` is being used to compute the cosine similarity between the items.

The code provided performs the computation of the cosine similarity between items. Note that the function `cosine_similarity` returns a Numpy data type and therefore the output has been converted to a Pandas data frame named `item_item_corr_matrix`. This variable contains the similarity values between items.

```
[11]: from google.colab import drive
      drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

```
[12]: #Exercise 4: Item-Item Filtering
#Authors: Ariton Verush

import numpy as np
import pandas as pd
from sklearn.metrics.pairwise import cosine_similarity

import math

*****MAIN
↳PROGRAM*****
# read data
#
movie_ratings = pd.read_csv('/content/movies_rating.csv', header=0,
↳index_col='Movie')

print(movie_ratings)

#Calculating the cosine similarity across items. Note that the
↳cosine_similarity function depends from the sklearn library
#As the cosine_similarity function returns an array, we create a new data frame
↳and assign the names to columns and rows
#which come from the original dataframe

name_movies= list(movie_ratings.index)

item_item_corr_matrix= pd.DataFrame(cosine_similarity(movie_ratings),
↳columns=name_movies, index=name_movies)
print("\nSimilarity Matrx: \n")
print(item_item_corr_matrix)
```

	Pedro	Pablo	Arianna	\
Movie				
Toy Story (1995)	2	1	0	
Star Wars: Episode VI - Return of the Jedi (1983)	5	0	0	
Forrest Gump (1994)	2	0	0	
Shawshank Redemption, The (1994)	0	2	5	
Silence of the Lambs, The (1991)	4	4	2	
Gladiator (2000)	4	2	0	
Star Wars: Episode IV - A New Hope (1977)	1	5	0	
Saving Private Ryan (1998)	2	0	0	
Pulp Fiction (1994)	0	0	0	
Stand by Me (1986)	3	4	1	
Shakespeare in Love (1998)	2	3	0	

Total Recall (1990)	0	2	1
Independence Day (ID4) (1996)	5	2	4
Blade Runner (1982)	2	0	4
Groundhog Day (1993)	5	2	1
Matrix, The (1999)	4	0	1
Schindler's List (1993)	2	5	2
Sixth Sense, The (1999)	5	1	3
Raiders of the Lost Ark (1981)	0	3	1
Babe (1995)	0	0	3

	Marcos	Arielle	Gio	\
Movie				
Toy Story (1995)	2	0	3	
Star Wars: Episode VI - Return of the Jedi (1983)	3	3	4	
Forrest Gump (1994)	0	4	4	
Shawshank Redemption, The (1994)	0	1	4	
Silence of the Lambs, The (1991)	0	4	0	
Gladiator (2000)	1	1	0	
Star Wars: Episode IV - A New Hope (1977)	0	4	2	
Saving Private Ryan (1998)	3	4	1	
Pulp Fiction (1994)	4	0	4	
Stand by Me (1986)	0	1	4	
Shakespeare in Love (1998)	0	5	5	
Total Recall (1990)	4	1	2	
Independence Day (ID4) (1996)	0	0	4	
Blade Runner (1982)	3	4	0	
Groundhog Day (1993)	5	0	1	
Matrix, The (1999)	0	3	0	
Schindler's List (1993)	5	5	0	
Sixth Sense, The (1999)	1	5	3	
Raiders of the Lost Ark (1981)	1	0	0	
Babe (1995)	2	0	2	

	Sofia	Moreno	Prisca	Tue	\
Movie					
Toy Story (1995)	0	2	4	4	
Star Wars: Episode VI - Return of the Jedi (1983)	0	0	1	2	
Forrest Gump (1994)	3	0	0	0	
Shawshank Redemption, The (1994)	1	0	4	5	
Silence of the Lambs, The (1991)	1	3	2	3	
Gladiator (2000)	4	2	4	0	
Star Wars: Episode IV - A New Hope (1977)	4	4	5	4	
Saving Private Ryan (1998)	5	0	4	3	
Pulp Fiction (1994)	2	3	0	4	
Stand by Me (1986)	0	0	1	0	
Shakespeare in Love (1998)	1	0	2	0	
Total Recall (1990)	0	2	0	0	
Independence Day (ID4) (1996)	0	1	1	2	

Blade Runner (1982)	3	2	0	0
Groundhog Day (1993)	0	4	5	0
Matrix, The (1999)	1	0	0	2
Schindler's List (1993)	0	1	0	5
Sixth Sense, The (1999)	0	3	0	1
Raiders of the Lost Ark (1981)	5	5	0	0
Babe (1995)	2	0	5	0

	Paula	Letty	James	Osito	\
Movie					
Toy Story (1995)	5	2	4	3	
Star Wars: Episode VI - Return of the Jedi (1983)	1	5	0	0	
Forrest Gump (1994)	5	2	0	3	
Shawshank Redemption, The (1994)	0	0	5	0	
Silence of the Lambs, The (1991)	0	0	4	2	
Gladiator (2000)	5	0	0	0	
Star Wars: Episode IV - A New Hope (1977)	2	3	2	3	
Saving Private Ryan (1998)	0	0	5	0	
Pulp Fiction (1994)	0	1	0	0	
Stand by Me (1986)	0	0	0	4	
Shakespeare in Love (1998)	0	3	0	0	
Total Recall (1990)	3	0	2	1	
Independence Day (ID4) (1996)	2	3	4	2	
Blade Runner (1982)	0	0	0	2	
Groundhog Day (1993)	0	2	3	3	
Matrix, The (1999)	1	5	0	5	
Schindler's List (1993)	0	0	0	1	
Sixth Sense, The (1999)	2	0	0	0	
Raiders of the Lost Ark (1981)	1	0	5	0	
Babe (1995)	4	2	0	0	

	Eva	Vanessa	Ginger	\
Movie				
Toy Story (1995)	2	0	2	
Star Wars: Episode VI - Return of the Jedi (1983)	4	0	0	
Forrest Gump (1994)	0	1	0	
Shawshank Redemption, The (1994)	0	0	0	
Silence of the Lambs, The (1991)	5	3	0	
Gladiator (2000)	5	2	0	
Star Wars: Episode IV - A New Hope (1977)	4	0	0	
Saving Private Ryan (1998)	0	2	0	
Pulp Fiction (1994)	3	0	2	
Stand by Me (1986)	5	1	0	
Shakespeare in Love (1998)	0	5	2	
Total Recall (1990)	1	0	0	
Independence Day (ID4) (1996)	3	2	3	
Blade Runner (1982)	0	0	0	
Groundhog Day (1993)	2	0	3	

Matrix, The (1999)	0	2	4
Schindler's List (1993)	3	2	0
Sixth Sense, The (1999)	0	4	0
Raiders of the Lost Ark (1981)	3	3	0
Babe (1995)	1	4	0

	Allan	Betty	Denisse
Movie			
Toy Story (1995)	4	2	2
Star Wars: Episode VI - Return of the Jedi (1983)	5	1	2
Forrest Gump (1994)	1	3	2
Shawshank Redemption, The (1994)	5	0	4
Silence of the Lambs, The (1991)	1	0	5
Gladiator (2000)	1	4	0
Star Wars: Episode IV - A New Hope (1977)	4	5	1
Saving Private Ryan (1998)	0	1	3
Pulp Fiction (1994)	5	3	2
Stand by Me (1986)	0	3	2
Shakespeare in Love (1998)	0	3	1
Total Recall (1990)	0	1	0
Independence Day (ID4) (1996)	0	0	4
Blade Runner (1982)	4	0	5
Groundhog Day (1993)	0	0	5
Matrix, The (1999)	0	2	4
Schindler's List (1993)	2	0	3
Sixth Sense, The (1999)	1	0	5
Raiders of the Lost Ark (1981)	2	0	3
Babe (1995)	5	0	0

Similarity Matrx:

	Toy Story (1995) \
Toy Story (1995)	1.000000
Star Wars: Episode VI - Return of the Jedi (1983)	0.644995
Forrest Gump (1994)	0.580540
Shawshank Redemption, The (1994)	0.667424
Silence of the Lambs, The (1991)	0.570229
Gladiator (2000)	0.587852
Star Wars: Episode IV - A New Hope (1977)	0.747409
Saving Private Ryan (1998)	0.534579
Pulp Fiction (1994)	0.667846
Stand by Me (1986)	0.492659
Shakespeare in Love (1998)	0.376659
Total Recall (1990)	0.623056
Independence Day (ID4) (1996)	0.690665
Blade Runner (1982)	0.383067
Groundhog Day (1993)	0.661016
Matrix, The (1999)	0.505010

Schindler's List (1993)	0.463817
Sixth Sense, The (1999)	0.421637
Raiders of the Lost Ark (1981)	0.466817
Babe (1995)	0.618070

Star Wars: Episode VI -

Return of the Jedi (1983) \	
Toy Story (1995)	
0.644995	
Star Wars: Episode VI - Return of the Jedi (1983)	
1.000000	
Forrest Gump (1994)	
0.563029	
Shawshank Redemption, The (1994)	
0.456052	
Silence of the Lambs, The (1991)	
0.516566	
Gladiator (2000)	
0.483187	
Star Wars: Episode IV - A New Hope (1977)	
0.589805	
Saving Private Ryan (1998)	
0.408752	
Pulp Fiction (1994)	
0.685662	
Stand by Me (1986)	
0.534324	
Shakespeare in Love (1998)	
0.533429	
Total Recall (1990)	
0.391934	
Independence Day (ID4) (1996)	
0.605856	
Blade Runner (1982)	
0.515397	
Groundhog Day (1993)	
0.526952	
Matrix, The (1999)	
0.535673	
Schindler's List (1993)	
0.573529	
Sixth Sense, The (1999)	
0.565297	
Raiders of the Lost Ark (1981)	
0.252604	
Babe (1995)	
0.511576	

	Forrest Gump (1994) \
Toy Story (1995)	0.580540
Star Wars: Episode VI - Return of the Jedi (1983)	0.563029
Forrest Gump (1994)	1.000000
Shawshank Redemption, The (1994)	0.293041
Silence of the Lambs, The (1991)	0.381346
Gladiator (2000)	0.569209
Star Wars: Episode IV - A New Hope (1977)	0.595550
Saving Private Ryan (1998)	0.463003
Pulp Fiction (1994)	0.399114
Stand by Me (1986)	0.527926
Shakespeare in Love (1998)	0.647153
Total Recall (1990)	0.491498
Independence Day (ID4) (1996)	0.498741
Blade Runner (1982)	0.487713
Groundhog Day (1993)	0.298290
Matrix, The (1999)	0.631039
Schindler's List (1993)	0.320494
Sixth Sense, The (1999)	0.602943
Raiders of the Lost Ark (1981)	0.288275
Babe (1995)	0.456849

Shawshank Redemption, The

(1994) \
Toy Story (1995)
0.667424
Star Wars: Episode VI - Return of the Jedi (1983)
0.456052
Forrest Gump (1994)
0.293041
Shawshank Redemption, The (1994)
1.000000
Silence of the Lambs, The (1991)
0.589000
Gladiator (2000)
0.212846
Star Wars: Episode IV - A New Hope (1977)
0.565577
Saving Private Ryan (1998)
0.598344
Pulp Fiction (1994)
0.538219
Stand by Me (1986)
0.340151
Shakespeare in Love (1998)
0.329203
Total Recall (1990)
0.332674

Independence Day (ID4) (1996)
 0.617366
 Blade Runner (1982)
 0.531981
 Groundhog Day (1993)
 0.437319
 Matrix, The (1999)
 0.255345
 Schindler's List (1993)
 0.497511
 Sixth Sense, The (1999)
 0.459446
 Raiders of the Lost Ark (1981)
 0.467347
 Babe (1995)
 0.542782

Silence of the Lambs, The

(1991) \
 Toy Story (1995)
 0.570229
 Star Wars: Episode VI - Return of the Jedi (1983)
 0.516566
 Forrest Gump (1994)
 0.381346
 Shawshank Redemption, The (1994)
 0.589000
 Silence of the Lambs, The (1991)
 1.000000
 Gladiator (2000)
 0.551612
 Star Wars: Episode IV - A New Hope (1977)
 0.682137
 Saving Private Ryan (1998)
 0.640590
 Pulp Fiction (1994)
 0.400471
 Stand by Me (1986)
 0.661958
 Shakespeare in Love (1998)
 0.484751
 Total Recall (1990)
 0.414499
 Independence Day (ID4) (1996)
 0.738445
 Blade Runner (1982)
 0.585662
 Groundhog Day (1993)

0.673091
 Matrix, The (1999)
 0.530856
 Schindler's List (1993)
 0.757630
 Sixth Sense, The (1999)
 0.715565
 Raiders of the Lost Ark (1981)
 0.702452
 Babe (1995)
 0.309159

	Gladiator (2000) \
Toy Story (1995)	0.587852
Star Wars: Episode VI - Return of the Jedi (1983)	0.483187
Forrest Gump (1994)	0.569209
Shawshank Redemption, The (1994)	0.212846
Silence of the Lambs, The (1991)	0.551612
Gladiator (2000)	1.000000
Star Wars: Episode IV - A New Hope (1977)	0.716828
Saving Private Ryan (1998)	0.476194
Pulp Fiction (1994)	0.414129
Stand by Me (1986)	0.566327
Shakespeare in Love (1998)	0.433263
Total Recall (1990)	0.480317
Independence Day (ID4) (1996)	0.442199
Blade Runner (1982)	0.303637
Groundhog Day (1993)	0.470793
Matrix, The (1999)	0.318849
Schindler's List (1993)	0.385040
Sixth Sense, The (1999)	0.415715
Raiders of the Lost Ark (1981)	0.526838
Babe (1995)	0.576106

	Star Wars: Episode IV - A New
Hope (1977) \	
Toy Story (1995)	
0.747409	
Star Wars: Episode VI - Return of the Jedi (1983)	
0.589805	
Forrest Gump (1994)	
0.595550	
Shawshank Redemption, The (1994)	
0.565577	
Silence of the Lambs, The (1991)	
0.682137	
Gladiator (2000)	
0.716828	

Star Wars: Episode IV - A New Hope (1977)
 1.000000
 Saving Private Ryan (1998)
 0.579057
 Pulp Fiction (1994)
 0.633846
 Stand by Me (1986)
 0.627804
 Shakespeare in Love (1998)
 0.560430
 Total Recall (1990)
 0.496723
 Independence Day (ID4) (1996)
 0.495896
 Blade Runner (1982)
 0.449518
 Groundhog Day (1993)
 0.515335
 Matrix, The (1999)
 0.470223
 Schindler's List (1993)
 0.583787
 Sixth Sense, The (1999)
 0.406425
 Raiders of the Lost Ark (1981)
 0.581505
 Babe (1995)
 0.506526

	Saving Private Ryan (1998) \
Toy Story (1995)	0.534579
Star Wars: Episode VI - Return of the Jedi (1983)	0.408752
Forrest Gump (1994)	0.463003
Shawshank Redemption, The (1994)	0.598344
Silence of the Lambs, The (1991)	0.640590
Gladiator (2000)	0.476194
Star Wars: Episode IV - A New Hope (1977)	0.579057
Saving Private Ryan (1998)	1.000000
Pulp Fiction (1994)	0.405308
Stand by Me (1986)	0.267182
Shakespeare in Love (1998)	0.493657
Total Recall (1990)	0.391963
Independence Day (ID4) (1996)	0.468207
Blade Runner (1982)	0.532917
Groundhog Day (1993)	0.556020
Matrix, The (1999)	0.406671
Schindler's List (1993)	0.526662
Sixth Sense, The (1999)	0.506329

Raiders of the Lost Ark (1981)	0.573845
Babe (1995)	0.405763

	Pulp Fiction (1994) \
Toy Story (1995)	0.667846
Star Wars: Episode VI - Return of the Jedi (1983)	0.685662
Forrest Gump (1994)	0.399114
Shawshank Redemption, The (1994)	0.538219
Silence of the Lambs, The (1991)	0.400471
Gladiator (2000)	0.414129
Star Wars: Episode IV - A New Hope (1977)	0.633846
Saving Private Ryan (1998)	0.405308
Pulp Fiction (1994)	1.000000
Stand by Me (1986)	0.416002
Shakespeare in Love (1998)	0.349375
Total Recall (1990)	0.499326
Independence Day (ID4) (1996)	0.424421
Blade Runner (1982)	0.500537
Groundhog Day (1993)	0.450466
Matrix, The (1999)	0.315125
Schindler's List (1993)	0.548530
Sixth Sense, The (1999)	0.368747
Raiders of the Lost Ark (1981)	0.467642
Babe (1995)	0.452605

	Stand by Me (1986) \
Toy Story (1995)	0.492659
Star Wars: Episode VI - Return of the Jedi (1983)	0.534324
Forrest Gump (1994)	0.527926
Shawshank Redemption, The (1994)	0.340151
Silence of the Lambs, The (1991)	0.661958
Gladiator (2000)	0.566327
Star Wars: Episode IV - A New Hope (1977)	0.627804
Saving Private Ryan (1998)	0.267182
Pulp Fiction (1994)	0.416002
Stand by Me (1986)	1.000000
Shakespeare in Love (1998)	0.569224
Total Recall (1990)	0.444554
Independence Day (ID4) (1996)	0.658770
Blade Runner (1982)	0.316894
Groundhog Day (1993)	0.521370
Matrix, The (1999)	0.473158
Schindler's List (1993)	0.517088
Sixth Sense, The (1999)	0.474540
Raiders of the Lost Ark (1981)	0.342329
Babe (1995)	0.241775

Shakespeare in Love (1998) \

Toy Story (1995)	0.376659
Star Wars: Episode VI - Return of the Jedi (1983)	0.533429
Forrest Gump (1994)	0.647153
Shawshank Redemption, The (1994)	0.329203
Silence of the Lambs, The (1991)	0.484751
Gladiator (2000)	0.433263
Star Wars: Episode IV - A New Hope (1977)	0.560430
Saving Private Ryan (1998)	0.493657
Pulp Fiction (1994)	0.349375
Stand by Me (1986)	0.569224
Shakespeare in Love (1998)	1.000000
Total Recall (1990)	0.328551
Independence Day (ID4) (1996)	0.529549
Blade Runner (1982)	0.292754
Groundhog Day (1993)	0.355682
Matrix, The (1999)	0.563204
Schindler's List (1993)	0.453812
Sixth Sense, The (1999)	0.645179
Raiders of the Lost Ark (1981)	0.273514
Babe (1995)	0.428845

	Total Recall (1990) \
Toy Story (1995)	0.623056
Star Wars: Episode VI - Return of the Jedi (1983)	0.391934
Forrest Gump (1994)	0.491498
Shawshank Redemption, The (1994)	0.332674
Silence of the Lambs, The (1991)	0.414499
Gladiator (2000)	0.480317
Star Wars: Episode IV - A New Hope (1977)	0.496723
Saving Private Ryan (1998)	0.391963
Pulp Fiction (1994)	0.499326
Stand by Me (1986)	0.444554
Shakespeare in Love (1998)	0.328551
Total Recall (1990)	1.000000
Independence Day (ID4) (1996)	0.464390
Blade Runner (1982)	0.377725
Groundhog Day (1993)	0.541289
Matrix, The (1999)	0.186883
Schindler's List (1993)	0.543651
Sixth Sense, The (1999)	0.420326
Raiders of the Lost Ark (1981)	0.502206
Babe (1995)	0.397253

	Independence Day (ID4) (1996) \
Toy Story (1995)	0.690665
Star Wars: Episode VI - Return of the Jedi (1983)	0.605856
Forrest Gump (1994)	0.498741

Shawshank Redemption, The (1994)	0.617366
Silence of the Lambs, The (1991)	0.738445
Gladiator (2000)	0.442199
Star Wars: Episode IV - A New Hope (1977)	0.495896
Saving Private Ryan (1998)	0.468207
Pulp Fiction (1994)	0.424421
Stand by Me (1986)	0.658770
Shakespeare in Love (1998)	0.529549
Total Recall (1990)	0.464390
Independence Day (ID4) (1996)	1.000000
Blade Runner (1982)	0.436159
Groundhog Day (1993)	0.713346
Matrix, The (1999)	0.670502
Schindler's List (1993)	0.481765
Sixth Sense, The (1999)	0.667356
Raiders of the Lost Ark (1981)	0.501533
Babe (1995)	0.409561

	Blade Runner (1982) \
Toy Story (1995)	0.383067
Star Wars: Episode VI - Return of the Jedi (1983)	0.515397
Forrest Gump (1994)	0.487713
Shawshank Redemption, The (1994)	0.531981
Silence of the Lambs, The (1991)	0.585662
Gladiator (2000)	0.303637
Star Wars: Episode IV - A New Hope (1977)	0.449518
Saving Private Ryan (1998)	0.532917
Pulp Fiction (1994)	0.500537
Stand by Me (1986)	0.316894
Shakespeare in Love (1998)	0.292754
Total Recall (1990)	0.377725
Independence Day (ID4) (1996)	0.436159
Blade Runner (1982)	1.000000
Groundhog Day (1993)	0.534737
Matrix, The (1999)	0.508483
Schindler's List (1993)	0.625236
Sixth Sense, The (1999)	0.702241
Raiders of the Lost Ark (1981)	0.498888
Babe (1995)	0.417179

	Groundhog Day (1993) \
Toy Story (1995)	0.661016
Star Wars: Episode VI - Return of the Jedi (1983)	0.526952
Forrest Gump (1994)	0.298290
Shawshank Redemption, The (1994)	0.437319
Silence of the Lambs, The (1991)	0.673091
Gladiator (2000)	0.470793
Star Wars: Episode IV - A New Hope (1977)	0.515335

Saving Private Ryan (1998)	0.556020
Pulp Fiction (1994)	0.450466
Stand by Me (1986)	0.521370
Shakespeare in Love (1998)	0.355682
Total Recall (1990)	0.541289
Independence Day (ID4) (1996)	0.713346
Blade Runner (1982)	0.534737
Groundhog Day (1993)	1.000000
Matrix, The (1999)	0.563592
Schindler's List (1993)	0.513265
Sixth Sense, The (1999)	0.533244
Raiders of the Lost Ark (1981)	0.499595
Babe (1995)	0.353261

Matrix, The (1999) \

Toy Story (1995)	0.505010
Star Wars: Episode VI - Return of the Jedi (1983)	0.535673
Forrest Gump (1994)	0.631039
Shawshank Redemption, The (1994)	0.255345
Silence of the Lambs, The (1991)	0.530856
Gladiator (2000)	0.318849
Star Wars: Episode IV - A New Hope (1977)	0.470223
Saving Private Ryan (1998)	0.406671
Pulp Fiction (1994)	0.315125
Stand by Me (1986)	0.473158
Shakespeare in Love (1998)	0.563204
Total Recall (1990)	0.186883
Independence Day (ID4) (1996)	0.670502
Blade Runner (1982)	0.508483
Groundhog Day (1993)	0.563592
Matrix, The (1999)	1.000000
Schindler's List (1993)	0.434749
Sixth Sense, The (1999)	0.564590
Raiders of the Lost Ark (1981)	0.208362
Babe (1995)	0.235219

Schindler's List (1993) \

Toy Story (1995)	0.463817
Star Wars: Episode VI - Return of the Jedi (1983)	0.573529
Forrest Gump (1994)	0.320494
Shawshank Redemption, The (1994)	0.497511
Silence of the Lambs, The (1991)	0.757630
Gladiator (2000)	0.385040
Star Wars: Episode IV - A New Hope (1977)	0.583787
Saving Private Ryan (1998)	0.526662
Pulp Fiction (1994)	0.548530
Stand by Me (1986)	0.517088
Shakespeare in Love (1998)	0.453812

Total Recall (1990)	0.543651
Independence Day (ID4) (1996)	0.481765
Blade Runner (1982)	0.625236
Groundhog Day (1993)	0.513265
Matrix, The (1999)	0.434749
Schindler's List (1993)	1.000000
Sixth Sense, The (1999)	0.641689
Raiders of the Lost Ark (1981)	0.434162
Babe (1995)	0.305295

	Sixth Sense, The (1999) \
Toy Story (1995)	0.421637
Star Wars: Episode VI - Return of the Jedi (1983)	0.565297
Forrest Gump (1994)	0.602943
Shawshank Redemption, The (1994)	0.459446
Silence of the Lambs, The (1991)	0.715565
Gladiator (2000)	0.415715
Star Wars: Episode IV - A New Hope (1977)	0.406425
Saving Private Ryan (1998)	0.506329
Pulp Fiction (1994)	0.368747
Stand by Me (1986)	0.474540
Shakespeare in Love (1998)	0.645179
Total Recall (1990)	0.420326
Independence Day (ID4) (1996)	0.667356
Blade Runner (1982)	0.702241
Groundhog Day (1993)	0.533244
Matrix, The (1999)	0.564590
Schindler's List (1993)	0.641689
Sixth Sense, The (1999)	1.000000
Raiders of the Lost Ark (1981)	0.434660
Babe (1995)	0.394331

	Raiders of the Lost Ark
(1981) \	
Toy Story (1995)	0.466817
Star Wars: Episode VI - Return of the Jedi (1983)	0.252604
Forrest Gump (1994)	0.288275
Shawshank Redemption, The (1994)	0.467347
Silence of the Lambs, The (1991)	0.702452
Gladiator (2000)	0.526838
Star Wars: Episode IV - A New Hope (1977)	0.581505

Saving Private Ryan (1998)
 0.573845
 Pulp Fiction (1994)
 0.467642
 Stand by Me (1986)
 0.342329
 Shakespeare in Love (1998)
 0.273514
 Total Recall (1990)
 0.502206
 Independence Day (ID4) (1996)
 0.501533
 Blade Runner (1982)
 0.498888
 Groundhog Day (1993)
 0.499595
 Matrix, The (1999)
 0.208362
 Schindler's List (1993)
 0.434162
 Sixth Sense, The (1999)
 0.434660
 Raiders of the Lost Ark (1981)
 1.000000
 Babe (1995)
 0.389762

	Babe (1995)
Toy Story (1995)	0.618070
Star Wars: Episode VI - Return of the Jedi (1983)	0.511576
Forrest Gump (1994)	0.456849
Shawshank Redemption, The (1994)	0.542782
Silence of the Lambs, The (1991)	0.309159
Gladiator (2000)	0.576106
Star Wars: Episode IV - A New Hope (1977)	0.506526
Saving Private Ryan (1998)	0.405763
Pulp Fiction (1994)	0.452605
Stand by Me (1986)	0.241775
Shakespeare in Love (1998)	0.428845
Total Recall (1990)	0.397253
Independence Day (ID4) (1996)	0.409561
Blade Runner (1982)	0.417179
Groundhog Day (1993)	0.353261
Matrix, The (1999)	0.235219
Schindler's List (1993)	0.305295
Sixth Sense, The (1999)	0.394331
Raiders of the Lost Ark (1981)	0.389762
Babe (1995)	1.000000

Provide the title of the five (5) most similar movies to Toy Story, in decreasing order of similarity.

```
[13]: #Write your code here
item_corr_top5 = (item_item_corr_matrix.loc['Toy Story (1995)'].
    ↪sort_values(ascending=0)[1:6])
print(item_corr_top5)
```

```
Star Wars: Episode IV - A New Hope (1977)    0.747409
Independence Day (ID4) (1996)                0.690665
Pulp Fiction (1994)                          0.667846
Shawshank Redemption, The (1994)             0.667424
Groundhog Day (1993)                         0.661016
Name: Toy Story (1995), dtype: float64
```

*Answer: The top 5 most similar movies to Toy Story are:

1. Star Wars: Episode IV - A New Hope (1977), score: 0.747409
2. Independence Day (ID4) (1996), score: 0.690665
3. Pulp Fiction (1994), score: 0.667846
4. Shawshank Redemption, The (1994), score: 0.667424
5. Groundhog Day (1993), score: 0.661016

3 Task 2: Compute a prediction score for target itms

Provide a prediction score for *all* the movies that user *Pablo* has not rated yet. To perform this task take into consideration the following:

- The rating for a movie that the user has not rated is equal to zero (0).
- You already have a similarity matrix with similarity values among the movies.
- Refer to the step 3 of the Instructions section. There you will find the formula that you need to use to obtain the prediction scores on unrated items. For the sake of this exercise, you will not select the most similar items, you will have to use all the similarity values of all the rated items by the user.

Hint: The prediction score for the movie Star Wars: Episode VI - Return of the Jedi (1983) is of 2.774

```
[14]: #Write your code here

predictions = {}
pablo_ratings = (movie_ratings['Pablo'].sort_values(ascending=0))

# movie loop, checking each one if pablo has rated or not and then making
↪prediction about the ones he hasnt rated
for i in range(0, len(pablo_ratings.index)):
    sum1 = 0;
    sum2 = 0;
    if pablo_ratings[i] == 0:
```

```

correlation = item_item_corr_matrix[pablo_ratings.index[i]]

for j in range(0, len(pablo_ratings.index)):
    if pablo_ratings[j] != 0:
        sum1 = sum1 + correlation[pablo_ratings.index[j]] * pablo_ratings[j]
        sum2 = sum2 + correlation[pablo_ratings.index[j]]

predictions[pablo_ratings.index[i]] = sum1 / sum2

predictions

```

```

[14]: {'Blade Runner (1982)': 2.7842520940793487,
      'Star Wars: Episode VI - Return of the Jedi (1983)': 2.774215215830409,
      'Matrix, The (1999)': 2.7701762031234276,
      'Pulp Fiction (1994)': 2.802797101190715,
      'Saving Private Ryan (1998)': 2.7802181220722764,
      'Forrest Gump (1994)': 2.708550273096228,
      'Babe (1995)': 2.6109454185869554}

```

*Answer: Predictions about the movies that Pablo hasn't rated yet:

Blade Runner (1982): 2.7842520940793487

Star Wars: Episode VI - Return of the Jedi (1983): 2.774215215830409

Matrix, The (1999): 2.7701762031234276

Pulp Fiction (1994): 2.802797101190715

Saving Private Ryan (1998): 2.7802181220722764

Forrest Gump (1994): 2.708550273096228

Babe (1995): 2.6109454185869554

4 Task 3: Answer the following

- Using the prediction scores computed in *Task 2*, which movie will you recommend to the user Pablo? > *Answer: Pulp Fiction (1994) has the highest rating of 2.8, however, I would also recommend Blade Runner (1982) and Saving Private Ryan (1998) coming on the second place with a prediction score of 2.78.
- What are the main differences between User-User and Item-Item collaborative filtering? > *Answer: In User-User collaborative filtering, similarities between users are calculated based on their interactions with items. On the other hand, in Item-Item collaborative filtering, similarities between items are calculated based on the interactions or ratings they have received from users.
- Under which circumstances will you prefer Item-Item collaborative filtering over User-User collaborative filtering? > *Answer: I would prefer Item-Item collaborative filtering over User-User in cases where we have a sparse user-item matrix, as well as scenarios where we have a

huge amount of items and less users. Also, it might be good to use this type of filtering for new users, since he/she is new and it is difficult to recommend items for such users.